

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented agentic workflows orchestrating **LLMs** (Claude, OpenAI) and **MCP clients/servers**, integrating with **GitHub Actions** for automated code reviews and PR comment bots.
- Built reusable **ReactJS** UI components for agent invocation and status visualization, ensuring adherence to **design systems** and **accessibility** standards.
- Developed **Node.js** services and **Python** scripts for agent backend integration, implementing **prompt engineering**, **tool schemas**, and **function calling** patterns for reliable agent execution.
- Engineered **Python** data pipelines on **AWS Lambda** to process and enrich data for AI agent context, handling approximately **3,500 events/sec** during peak hours.
- Implemented prompt context management, including **RAG pipelines** and lightweight vector stores, to augment agent memory and retrieval strategies, improving response relevance by an estimated **15%**.
- Shipped high-quality code for agent workflows and UI components, ensuring coverage with **unit and e2e tests** (Jest/Cypress), integrated into **CI/CD pipelines** with comprehensive **observability** (logs, metrics, traces).
- Documented system architecture and **OpenAI/MCP spec contracts**, creating developer guides for internal consumers to facilitate integration and adoption of AI agent tooling.

SKILLS

AI/ML – LLMs, OpenAI, Claude, MCP, Agentic Workflows, Prompt Engineering, Function Calling, RAG, Embeddings, Vector Stores, OpenAI Specifications

Frontend – ReactJS, AngularJS, TypeScript, JavaScript, HTML, CSS, Accessibility, Design Systems

Backend – Node.js, Python, FastAPI, Flask, REST APIs, GraphQL

Cloud/DevOps – AWS, Lambda, EC2, S3, CloudWatch, Docker, Kubernetes, CI/CD, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle, SQL

Observability – Splunk, Dynatrace, Datadog, OpenTelemetry, Logging, Metrics, Tracing

PROJECTS

AI Agent Integration Platform (HCL Technologies)

- Developed backend services in **Python** and **Node.js** to integrate **LLM** capabilities into enterprise workflows, focusing on tool orchestration and function calling.
- Built a **ReactJS** frontend component library for agents, enabling visualization of execution status and error handling for a diverse set of internal users.
- Implemented **RAG pipelines** using **PostgreSQL** as a vector store for a pilot application, improving retrieval accuracy for agent-driven knowledge base queries by **~18%**.

Developer Productivity Agent (HCL Technologies)

- Engineered a **Python**-based agent that integrates with **GitHub APIs** to automate code review feedback and comment generation on pull requests, reducing manual review time by an estimated **20%**.
- Designed and implemented **OpenAI specification** contracts for agent tools, including input validation and schema definitions, to ensure consistent interaction with downstream services.
- Containerized the agent service using **Docker** and deployed it to an **AWS EC2** instance, establishing a CI/CD pipeline with **Jenkins** for automated testing and deployment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)